

Answer Key — Biology Grade 11

Sub-Strand 2.2: Growth and Development in Plants • Lesson 9: Role of Hormones in Plant Growth and Development: Why Do Maize and Bean Plants Grow Differently?

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7
A	A	D	B	C	A	A

USE AFTER: PREDICT PHASE — Predict: Think-Pair-Share on pinching off the tip of a young bean seedling

Q1. A farmer pinches off the very top (the shoot tip) of a young bean seedling. What is the most likely result after two weeks?

- A. More side branches grow because the tip no longer holds back the side buds**
- B. The plant dies because the tip is cut off
- C. Nothing changes because plants grow only from water and soil
- D. The plant grows taller than before because it has less to feed

Correct answer: A. The shoot tip makes auxin, which suppresses side buds. When the tip is removed, the side buds are released and the plant branches.

USE AFTER: OBSERVE PHASE — Observe: 'Plant Hormones' reading card on abscisic acid and dormancy

Q2. In the dry season, healthy seeds in the soil do not germinate even though they are not dead. Which plant hormone helps to keep these seeds dormant?

- A. Absciscic acid**
- B. Cytokinins
- C. Gibberellins
- D. Ethylene

Correct answer: A. Absciscic acid maintains dormancy and stops germination during unfavourable conditions. Gibberellins do the opposite and promote germination.

USE AFTER: OBSERVE PHASE — Observe: pruning demonstration of pruned versus intact bean seedlings

Q3. Two bean seedlings were grown for two weeks. The intact seedling was 30 cm tall with 0 side branches. The seedling with its tip pinched off was 18 cm tall with 4 side branches. Which conclusion is best supported by these results?

- A. Removing the tip kills the plant, so it cannot grow taller
- B. Hormones are found only in animals, so the plants differ by chance
- C. Side branches grow only when a plant is shorter than 20 cm
- D. Removing the shoot tip allowed side buds to grow, so the tip normally suppresses branching**

Correct answer: D. The pruned plant is alive and has more side branches, while the intact plant is taller with no branches. This shows the tip controls side bud growth.

USE AFTER: OBSERVE PHASE — Observe: 'Plant Hormones' reading card on ethylene and ripening bananas and mangoes

Q4. Traders in Mombasa put green bananas in a closed sack and they ripen faster. Which explanation is correct?

- A. Absciscic acid is released and speeds up the ripening
- B. Ethylene gas given off by the fruit collects in the sack and speeds up ripening**
- C. Auxin builds up in the sack and makes the fruit softer
- D. Cytokinins from the sack make the fruit cells divide faster

Correct answer: B. Ethylene is the hormone that ripens fruit. A closed sack traps the gas around the bananas so it has a stronger effect.

USE AFTER: EXPLAIN PHASE — Explain: Claim-Evidence-Reasoning on hormones and apical dominance

Q5. What does the term 'apical dominance' mean?

- A. The shoot tip bends towards light because of gibberellins
- B. The side buds always grow faster than the main shoot
- C. The shoot tip makes auxin that holds back the growth of side buds**
- D. The root tip controls how fast the shoot grows

Correct answer: C. Apical means at the top and dominance means control. Auxin from the shoot tip suppresses the side buds, so the plant keeps a single main stem.

USE AFTER: EXPLAIN PHASE — Explain: farm applications of pruning tea in Kericho

Q6. Tea farmers in Kericho prune their tea bushes regularly. How does pruning help?

- A. It removes the shoot tips, which reduces apical dominance so side buds grow and the bush becomes bushy**
- B. It makes the bush grow taller with one thick stem
- C. It adds abscisic acid to the bush so that it stops growing
- D. It kills the old shoots so that only roots keep growing

Correct answer: A. Cutting off the tips removes the source of auxin that suppresses side buds. The side buds then grow, giving more shoots to pluck.

USE AFTER: MODEL BUILDING PHASE — Model: revising the Day 1 model by adding hormone arrows and labels

Q7. A learner revises their Day 1 model of why maize and bean seeds behave differently. Which addition is correct?

- A. Gibberellins promote germination, but the seed also needs water, oxygen and a suitable temperature to respond**
- B. Hormones alone decide germination, so water and temperature can be removed from the model
- C. Each hormone does only one job, so no arrows should link hormones to each other
- D. Only animals have hormones, so the model should show only seed structure

Correct answer: A. Hormones work together with each other and with the earlier factors such as water, oxygen, temperature and dormancy. Hormones are not the only factor in germination.